Title: College Lost & Found Management System Using Singly Linked List

1. Problem Statement

To develop a College Lost & Found Management System using a Singly Linked List in C++. The system stores details of lost and found items such as Item ID, Item Name, and Status (Lost/Found). It allows the user to add, search, remove, and display item details dynamically using linked list operations.

2. Objectives

- To understand the concept of Singly Linked List.
- To implement dynamic memory allocation in C++.
- To add and store lost and found item details dynamically.
- To search for an item using its Item ID.
- To remove an item from the linked list.
- To display item details and their status efficiently.
- To understand the practical application of linked lists in college data management.
- To improve programming and problem-solving skills.

3. Algorithm

Step 1: Start the program.

Step 2: Define a structure Node containing Item ID, Item Name, Status (Lost/Found), and a pointer named next.

Step 3: Initialize head = NULL.

Step 4: Display a menu with options to add, search, remove, display items, and exit.

Step 5: To add an item, create a new node dynamically and accept its ID, name, and status from the user.

Step 6: If the linked list is empty, make the new node the head; otherwise, traverse the list and add the new node at the end.

Step 7: To search for an item, traverse the list using Item ID and display the details if the item is found.

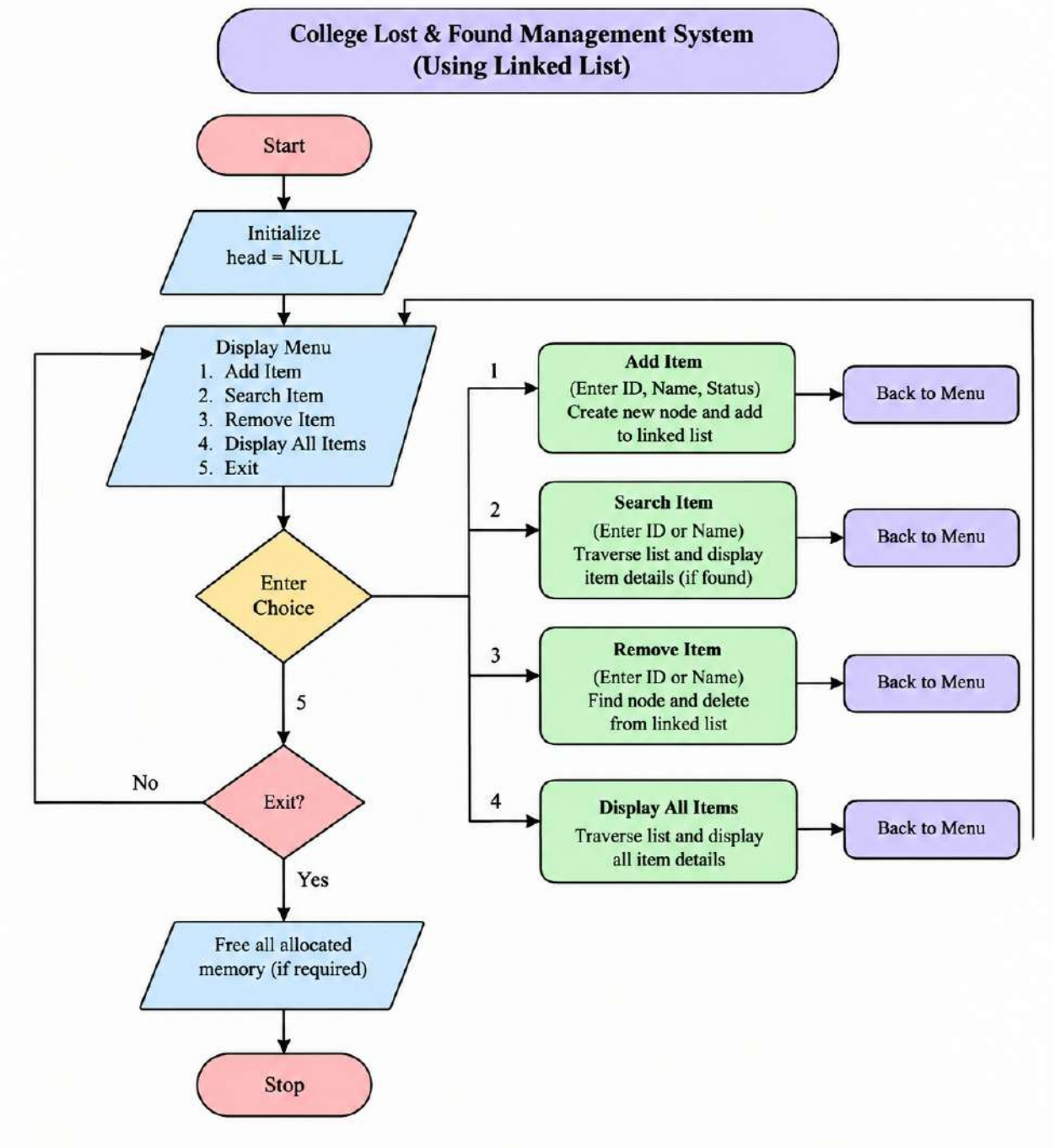
Step 8: To remove an item, find its node, adjust the links, and delete the node from memory.

Step 9: To display items, start from head, traverse the list, and display each item's details.

Step 10: Repeat the menu until the user selects Exit.

Step 11: Stop the program.

4. Flowchart


College Lost & Found Management System
(Using Linked List)
Start
Initialize
head = NULL
Display Menu
1. Add Item
2. Search Item
3. Remove Item
4. Display All Items
5. Exit
Enter
Choice
1
Add Item
(Enter ID, Name, Status)
Create new node and add
to linked list
Back to Menu
2
Search Item
(Enter ID or Name)
Traverse list and display
item details (if found)
Back to Menu
3
Remove Item
(Enter ID or Name)
Find node and delete
from linked list
Back to Menu
4
Display All Items
Traverse list and display
all item details
Back to Menu
5
Exit?
No
Yes
Free all allocated
memory (if required)
Stop

5. Conclusion

The College Lost & Found Management System using a Singly Linked List demonstrates the practical implementation of linked list concepts in C++. It provides a simple and dynamic method for adding, searching, removing, and displaying lost and found item records. This project helps in understanding nodes, pointers, dynamic memory allocation, traversal, and linked list operations. It also shows how linked lists can be used to manage item information in a college environment and improves practical programming and problem-solving skills.